

Qualification Pack



Rubber Internal Mixer Operator

QP Code: RSC/Q0112

Version: 4.0

NSQF Level: 4

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RSC/Q0112: Rubber Internal Mixer Operator

Brief Job Description

The Rubber Internal Mixer Operator is responsible for preparing, operating, and monitoring internal mixing equipment used in the processing of rubber compounds. The individual ensures accurate raw material loading, optimal machine operation, and adherence to process parameters to achieve consistent product quality. The operator also monitors the quality of the output mix, performs routine maintenance, and ensures workplace safety and compliance.

Personal Attributes

This job requires the individual to work independently and be comfortable in performing laborious work. He should be result oriented and positive in attitude. The individual must be willing to work in the factory environment.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [RSC/N9501: Prepare for Rubber Mixing Operations](#)
2. [RSC/N9502: Load Raw Materials, Operate and Control Automated Internal Mixer](#)
3. [RSC/N9503: Monitor Product and Process Quality During Mixing](#)
4. [RSC/N5007: Follow health & safety protocol](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Rubber
Sub-Sector	Industrial and Consumer Products
Occupation	industrial and consumer products production
Country	India
NSQF Level	4
Credits	17

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/8131.6200
Minimum Educational Qualification & Experience	12th grade Pass (Or Equivalent) OR 10th grade pass plus 2-year NTC (Or NAC Or CITS Or Equivalent) OR 10th grade pass with 3 Years of experience Relevant OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience Relevant
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	30/04/2025
Version	4.0
Reference code on NQR	QG-04-RI-04035-2025-V2-RSDC
NQR Version	2

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RSC/N9501: Prepare for Rubber Mixing Operations

Description

This NOS covers the competencies required to prepare the work area, tools, equipment, and raw materials before initiating internal rubber mixing operations.

Scope

The scope covers the following :

- Plan and Prepare Work Area
- Arrange Materials and Tools
- Prepare documents and specification
- Conduct pre-checks on mixer
- Follow safety and housekeeping protocols

Elements and Performance Criteria

Plan and prepare work area

To be competent, the user/individual on the job must be able to:

- PC1.** inspect the cleanliness and safety of the work area before beginning operations
- PC2.** ensure availability of safety gear and personal protective equipment (PPE)
- PC3.** follow safety protocols while entering mixing area
- PC4.** remove any leftover materials or obstructions from the mixing area

Arrange materials and tools

To be competent, the user/individual on the job must be able to:

- PC5.** check the availability of raw materials as per mixing plan
- PC6.** identify and label materials appropriately
- PC7.** ensure tools and containers are clean and in good condition
- PC8.** verify weighing balance calibration

Prepare documents and specifications

To be competent, the user/individual on the job must be able to:

- PC9.** read batch specifications and production plans
- PC10.** record and verify raw material batch numbers
- PC11.** maintain checklists and pre-operation logbook entries
- PC12.** confirm mixing schedule and shift-wise production targets

Conduct pre-checks on mixer

To be competent, the user/individual on the job must be able to:

- PC13.** inspect machine for any signs of damage or wear
- PC14.** ensure emergency stop buttons are functional
- PC15.** verify electrical supply and lubrication levels
- PC16.** check the temperature control systems and timer settings

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Follow safety and housekeeping protocols

To be competent, the user/individual on the job must be able to:

- PC17.** wear appropriate PPE during preparation activities
- PC18.** keep the floor area dry and free of slipping hazards
- PC19.** dispose of waste and contaminants as per SOP
- PC20.** follow standard housekeeping practices

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basics of rubber materials and types used in mixing
- KU2.** purpose and functioning of internal mixers
- KU3.** common tools and equipment used in rubber mixing
- KU4.** procedures for cleaning and setting up the work area
- KU5.** use and importance of batch sheets and logbooks
- KU6.** pre-operational checklist requirements
- KU7.** importance of PPE and safety practices
- KU8.** environmental regulations and waste handling
- KU9.** emergency procedures and fire safety
- KU10.** overview of hazards associated with raw rubber materials
- KU11.** basics of lubrication systems
- KU12.** importance of equipment calibration
- KU13.** role of coordination with other departments
- KU14.** standard housekeeping methods in rubber mixing units
- KU15.** common defects due to preparation errors

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret batch specifications and checklists
- GS2.** fill up logbooks, reports, and material labels
- GS3.** communicate issues with supervisors and coworkers clearly
- GS4.** follow numeric details like weights and ratios accurately
- GS5.** use basic arithmetic for calculating batch quantities
- GS6.** demonstrate discipline, cleanliness, and punctuality
- GS7.** work in a team to coordinate material availability
- GS8.** follow safety and emergency protocols

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Plan and prepare work area</i>	8	12	-	-
PC1. inspect the cleanliness and safety of the work area before beginning operations	2	3	-	-
PC2. ensure availability of safety gear and personal protective equipment (PPE)	2	3	-	-
PC3. follow safety protocols while entering mixing area	2	3	-	-
PC4. remove any leftover materials or obstructions from the mixing area	2	3	-	-
<i>Arrange materials and tools</i>	8	12	-	-
PC5. check the availability of raw materials as per mixing plan	2	3	-	-
PC6. identify and label materials appropriately	2	3	-	-
PC7. ensure tools and containers are clean and in good condition	2	3	-	-
PC8. verify weighing balance calibration	2	3	-	-
<i>Prepare documents and specifications</i>	8	12	-	-
PC9. read batch specifications and production plans	2	3	-	-
PC10. record and verify raw material batch numbers	2	3	-	-
PC11. maintain checklists and pre-operation logbook entries	2	3	-	-
PC12. confirm mixing schedule and shift-wise production targets	2	3	-	-
<i>Conduct pre-checks on mixer</i>	8	12	-	-
PC13. inspect machine for any signs of damage or wear	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. ensure emergency stop buttons are functional	2	3	-	-
PC15. verify electrical supply and lubrication levels	2	3	-	-
PC16. check the temperature control systems and timer settings	2	3	-	-
<i>Follow safety and housekeeping protocols</i>	8	12	-	-
PC17. wear appropriate PPE during preparation activities	2	3	-	-
PC18. keep the floor area dry and free of slipping hazards	2	3	-	-
PC19. dispose of waste and contaminants as per SOP	2	3	-	-
PC20. follow standard housekeeping practices	2	3	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N9501
NOS Name	Prepare for Rubber Mixing Operations
Sector	Rubber
Sub-Sector	
Occupation	industrial and consumer products production, Mixing
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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RSC/N9502: Load Raw Materials, Operate and Control Automated Internal Mixer

Description

This NOS covers the loading of raw materials into the mixer, operating the internal mixing machine, and controlling process parameters like temperature, pressure, and rotor speed. It includes monitoring automation systems and addressing in-process deviations.

Scope

The scope covers the following :

- Load raw materials and verify batch data
- Set and monitor machine parameters
- Handle machine interruptions and alarms
- Ensure product discharge and transfer
- Maintain process log and documentation

Elements and Performance Criteria

Load raw materials and verify batch data

To be competent, the user/individual on the job must be able to:

- PC1.** verify material identity and batch number before loading
- PC2.** sequence raw material addition as per job specification
- PC3.** ensure feeding port is clean and unobstructed
- PC4.** load rubber, fillers, and oils using safe material handling practices
- PC5.** use hoist or conveyor system where applicable
- PC6.** ensure correct dosing of accelerators and other chemicals

Set and monitor machine parameters

To be competent, the user/individual on the job must be able to:

- PC7.** input temperature, rotor RPM, and mixing time on control panel
- PC8.** start mixer and monitor control panel display
- PC9.** check torque value and power consumption indicators
- PC10.** monitor ram pressure and chamber temperature
- PC11.** adjust water cooling cycle if overheating is observed
- PC12.** track real-time data on SCADA system or PLC panel

Handle machine interruptions and alarms

To be competent, the user/individual on the job must be able to:

- PC13.** recognize fault alarms and take corrective actions
- PC14.** pause operations in case of material overload or jamming
- PC15.** restart machine after resolving error as per SOP
- PC16.** inform maintenance team for recurring issues

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Ensure product discharge and transfer

To be competent, the user/individual on the job must be able to:

- PC17.** verify batch completion via alarm or timer
- PC18.** operate discharge gate and transfer mix to drop mill or conveyor
- PC19.** avoid material spillage and ensure safe handling
- PC20.** label batch container with date and compound ID

Maintain process log and documentation

To be competent, the user/individual on the job must be able to:

- PC21.** record temperature, time, and batch size
- PC22.** update daily production report in manual or digital format
- PC23.** escalate any deviations in quality or machine function

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** functions of various parts of an internal mixer
- KU2.** principles of batch mixing and dispersion
- KU3.** process parameters like fill factor, rotor speed, and time
- KU4.** temperature control and its impact on compound quality
- KU5.** types of automation and SCADA/PLC systems used in mixers
- KU6.** safety interlocks and auto shutdown mechanisms
- KU7.** common alarms and troubleshooting techniques
- KU8.** role of masterbatch and curing agents
- KU9.** mixing cycle for different compound formulations
- KU10.** hazard associated with overloading or overheating
- KU11.** importance of mixing consistency and homogeneity
- KU12.** procedures to handle minor and major breakdowns
- KU13.** quality checks during in-process mixing
- KU14.** importance of batch labeling and traceability
- KU15.** documentation formats and shift reports
- KU16.** significance of preventive maintenance during operation
- KU17.** procedures for emergency stop and evacuation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret SCADA/PLC panels
- GS2.** write down operational details in logbooks accurately
- GS3.** communicate with the control room or team for process alerts
- GS4.** apply mathematical reasoning to modify rotor RPM and batch timings

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- GS5.** report anomalies in torque and power values
- GS6.** follow structured problem-solving for alarm rectification
- GS7.** make quick decisions in case of equipment failure
- GS8.** prioritize safety while handling hot or reactive batches

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Load raw materials and verify batch data</i>	11	17	-	-
PC1. verify material identity and batch number before loading	2	3	-	-
PC2. sequence raw material addition as per job specification	2	3	-	-
PC3. ensure feeding port is clean and unobstructed	2	3	-	-
PC4. load rubber, fillers, and oils using safe material handling practices	2	3	-	-
PC5. use hoist or conveyor system where applicable	1	2	-	-
PC6. ensure correct dosing of accelerators and other chemicals	2	3	-	-
<i>Set and monitor machine parameters</i>	12	18	-	-
PC7. input temperature, rotor RPM, and mixing time on control panel	2	3	-	-
PC8. start mixer and monitor control panel display	2	3	-	-
PC9. check torque value and power consumption indicators	2	3	-	-
PC10. monitor ram pressure and chamber temperature	2	3	-	-
PC11. adjust water cooling cycle if overheating is observed	2	3	-	-
PC12. track real-time data on SCADA system or PLC panel	2	3	-	-
<i>Handle machine interruptions and alarms</i>	6	10	-	-
PC13. recognize fault alarms and take corrective actions	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. pause operations in case of material overload or jamming	2	3	-	-
PC15. restart machine after resolving error as per SOP	1	2	-	-
PC16. inform maintenance team for recurring issues	1	2	-	-
<i>Ensure product discharge and transfer</i>	6	10	-	-
PC17. verify batch completion via alarm or timer	1	2	-	-
PC18. operate discharge gate and transfer mix to drop mill or conveyor	2	3	-	-
PC19. avoid material spillage and ensure safe handling	2	3	-	-
PC20. label batch container with date and compound ID	1	2	-	-
<i>Maintain process log and documentation</i>	5	5	-	-
PC21. record temperature, time, and batch size	2	2	-	-
PC22. update daily production report in manual or digital format	2	2	-	-
PC23. escalate any deviations in quality or machine function	1	1	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N9502
NOS Name	Load Raw Materials, Operate and Control Automated Internal Mixer
Sector	Rubber
Sub-Sector	
Occupation	industrial and consumer products production, Mixing
NSQF Level	4
Credits	6
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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RSC/N9503: Monitor Product and Process Quality During Mixing

Description

This NOS deals with inspecting rubber batches for conformity, conducting basic in-process quality tests, documenting results, and coordinating with the QC department. It includes identifying non-conformities and initiating corrective measures.

Scope

The scope covers the following :

- Conduct in-process inspection
- Identify deviations and take action
- Maintain quality documentation
- Ensure workplace cleanliness and safety

Elements and Performance Criteria

Conduct in-process inspection

To be competent, the user/individual on the job must be able to:

- PC1.** collect sample from every batch post-discharge
- PC2.** check for color, texture, and lump formation
- PC3.** measure Mooney viscosity using testing equipment
- PC4.** perform specific gravity or density tests
- PC5.** test cure time and tensile strength where required

Identify deviations and take action

To be competent, the user/individual on the job must be able to:

- PC6.** compare sample properties with formulation specs
- PC7.** report off-spec batches to supervisor or QC
- PC8.** reject batch if contamination or improper mixing observed
- PC9.** adjust mixer parameters for the next batch accordingly

Maintain quality documentation

To be competent, the user/individual on the job must be able to:

- PC10.** record test results in batch inspection sheet
- PC11.** label sample containers with batch info and date
- PC12.** update process deviation records with corrective actions
- PC13.** maintain inspection logs and communicate with QC

Ensure workplace cleanliness and safety

To be competent, the user/individual on the job must be able to:

- PC14.** clean sampling tools after use
- PC15.** dispose of waste sample material as per SOP
- PC16.** follow safety norms while handling heated or uncured rubber

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basics of rubber chemistry and compound behavior
- KU2.** common quality parameters for rubber batches
- KU3.** role of viscosity, density, and tensile strength tests
- KU4.** significance of color and texture uniformity
- KU5.** procedure for sample collection and handling
- KU6.** operation of Mooney viscometer and other QC tools
- KU7.** types of product non-conformities and probable causes
- KU8.** rejection criteria and escalation protocol
- KU9.** internal and external documentation standards
- KU10.** SOPs for maintaining sampling tools
- KU11.** communication flow between production and QC teams
- KU12.** safety measures during sample handling
- KU13.** waste segregation and hazardous sample disposal
- KU14.** record keeping practices and traceability norms

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read test reports, specifications, and work instructions
- GS2.** write inspection results with clarity and accuracy
- GS3.** communicate with QC team for feedback and corrections
- GS4.** apply critical thinking for quality deviation analysis
- GS5.** use basic computer systems for report entry
- GS6.** prioritize tasks to meet inspection timelines
- GS7.** follow safety and documentation standards

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct in-process inspection</i>	16	21	-	-
PC1. collect sample from every batch post-discharge	3	4	-	-
PC2. check for color, texture, and lump formation	3	4	-	-
PC3. measure Mooney viscosity using testing equipment	4	5	-	-
PC4. perform specific gravity or density tests	3	4	-	-
PC5. test cure time and tensile strength where required	3	4	-	-
<i>Identify deviations and take action</i>	9	16	-	-
PC6. compare sample properties with formulation specs	2	4	-	-
PC7. report off-spec batches to supervisor or QC	2	4	-	-
PC8. reject batch if contamination or improper mixing observed	2	4	-	-
PC9. adjust mixer parameters for the next batch accordingly	3	4	-	-
<i>Maintain quality documentation</i>	9	14	-	-
PC10. record test results in batch inspection sheet	2	4	-	-
PC11. label sample containers with batch info and date	2	3	-	-
PC12. update process deviation records with corrective actions	2	3	-	-
PC13. maintain inspection logs and communicate with QC	3	4	-	-
<i>Ensure workplace cleanliness and safety</i>	6	9	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. clean sampling tools after use	2	3	-	-
PC15. dispose of waste sample material as per SOP	2	3	-	-
PC16. follow safety norms while handling heated or uncured rubber	2	3	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N9503
NOS Name	Monitor Product and Process Quality During Mixing
Sector	Rubber
Sub-Sector	
Occupation	industrial and consumer products production, Mixing
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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RSC/N5007: Follow health & safety protocol

Description

This unit is about maintaining health and safety of self and others at workplace.

Scope

The scope covers the following :

- Maintain a clean and efficient workplace
- Render appropriate emergency procedures
- Maintain standard safety procedures at the workplace
- Participate in safety awareness campaigns
- Understand potential sources of accidents
- Use safety gears to avoid accidents

Elements and Performance Criteria

Maintain a clean and efficient workplace

To be competent, the user/individual on the job must be able to:

- PC1.** undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor
- PC2.** work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy
- PC3.** read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc
- PC4.** prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.
- PC5.** use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** dispose off waste safely and correctly in a designated area
- PC7.** risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace
- PC8.** perform work in a manner which minimizes environmental damage
- PC9.** all procedures and work instructions for controlling risk are followed closely
- PC10.** report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger

Render appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC12.** follow emergency procedures as per company standards and workplace requirements

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- PC13.** use emergency equipment in accordance with manufacturers' specifications and workplace requirements
- PC14.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.
- PC15.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC16.** dispose off medical waste in accordance with workplace requirements
- PC17.** report details of first aid administered in accordance with work place procedures

Maintain standard safety procedures at the workplace

To be competent, the user/individual on the job must be able to:

- PC18.** comply with general safety procedures
- PC19.** follow standard safety procedures while handling equipment, hazardous material or tool
- PC20.** check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.
- PC21.** ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure
- PC22.** keep the workplace organized, swept, clean and hazard free

Participate in safety awareness campaigns

To be competent, the user/individual on the job must be able to:

- PC23.** attend fire drills and other safety related workshops organized at the workplace
- PC24.** be aware of first aid, evacuation and emergency procedures
- PC25.** be alert of any events and do not be negligent to any safety procedures to be followed

Understand potential sources of accidents

To be competent, the user/individual on the job must be able to:

- PC26.** avoid accidents while using hazardous chemicals, machines, sharp tools and equipment

Use safety gears to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC27.** use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)
- PC28.** handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies on incentives, delivery standards, and personnel management.
- KU2.** occupational safety and health policy followed
- KU3.** emergency evacuation procedure
- KU4.** medical policy
- KU5.** company laws and acts
- KU6.** the risks to health and safety and the measures to be taken to control those risks in the area of work

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- KU7.** workplace procedures and requirements for the handling of workplace injuries/illnesses.
- KU8.** basic emergency first aid procedure
- KU9.** local emergency services
- KU10.** reporting on accidents, incidents and problems to appropriate authorities.
- KU11.** how to use machines as per standard operating procedure
- KU12.** how to maintain work area safe and secure
- KU13.** use of hazardous materials, tools and equipments
- KU14.** emergency evacuation and first aid procedures to be followed
- KU15.** personal hygiene and fitness requirements
- KU16.** general duties under the relevant health and safety legislation
- KU17.** what personal protective equipment and clothing should be worn and how it is cared for
- KU18.** the correct and safe way to use materials and equipment required for work
- KU19.** the importance of good housekeeping in the workplace
- KU20.** safe disposal methods for waste
- KU21.** methods for minimizing environmental damage during work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data which are required for record keeping purpose
- GS2.** report problems to the appropriate person in a timely manner
- GS3.** write descriptions and details about incidents in reports
- GS4.** read instruction manuals for hand tools and equipment
- GS5.** read instructions on work orders and procedures
- GS6.** receive instructions and seek advice from superiors
- GS7.** communicate clearly and effectively with others
- GS8.** take a decision for any change/issue based on earlier successes(documented previous history)on similar issues
- GS9.** work out changes in case a new improved machine/equipment is added in the process or any new material/chemical is developed replacing existing one.
- GS10.** make changes in cycle time due to improved process.
- GS11.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management
- GS12.** consult the peer group and superiors to arrive at a favourable decision.
- GS13.** use of standard available problem solving techniques for decision making
- GS14.** review and analyze the process steps to check on system non adherence andnon conformity
- GS15.** review the current sop and other standards for continuous improvement to facilitate decision making
- GS16.** take a calculated risk with minimum losses
- GS17.** schedule daily activities and drawing up priorities; allocate start times, estimation of completion times and materials, equipment and assistance required for completion.

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- GS18.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)
- GS19.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS20.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS21.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS22.** work towards fulfilling the customers requirement as per their demand.
- GS23.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level
- GS24.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS25.** maintain good/cordial relation with customers.
- GS26.** work on the feedback received from customer regarding the product.
- GS27.** use first aid treatment in case of any injury/accident.
- GS28.** monitor and maintain the condition of tools and equipment
- GS29.** assess situation & identify appropriate control measures
- GS30.** act, communicate and report in emergency situation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain a clean and efficient workplace</i>	9	21	-	-
PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor	2	2	-	-
PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy	2	2	-	-
PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc	-	2	-	-
PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.	2	2	-	-
PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use	1	2	-	-
PC6. dispose off waste safely and correctly in a designated area	2	4	-	-
PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace	-	2	-	-
PC8. perform work in a manner which minimizes environmental damage	-	2	-	-
PC9. all procedures and work instructions for controlling risk are followed closely	-	1	-	-
PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger	-	2	-	-
<i>Render appropriate emergency procedures</i>	9	18	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	4	-	-
PC12. follow emergency procedures as per company standards and workplace requirements	2	4	-	-
PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements	2	4	-	-
PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	-	1	-	-
PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	2	-	-
PC16. dispose off medical waste in accordance with workplace requirements	-	1	-	-
PC17. report details of first aid administered in accordance with work place procedures	3	2	-	-
<i>Maintain standard safety procedures at the workplace</i>	6	15	-	-
PC18. comply with general safety procedures	2	4	-	-
PC19. follow standard safety procedures while handling equipment, hazardous material or tool	-	2	-	-
PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.	2	4	-	-
PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure	-	1	-	-
PC22. keep the workplace organized, swept, clean and hazard free	2	4	-	-
<i>Participate in safety awareness campaigns</i>	2	8	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. attend fire drills and other safety related workshops organized at the workplace	1	3	-	-
PC24. be aware of first aid, evacuation and emergency procedures	1	3	-	-
PC25. be alert of any events and do not be negligent to any safety procedures to be followed	-	2	-	-
<i>Understand potential sources of accidents</i>	1	3	-	-
PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment	1	3	-	-
<i>Use safety gears to avoid accidents</i>	3	5	-	-
PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)	2	2	-	-
PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders	1	3	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5007
NOS Name	Follow health & safety protocol
Sector	Rubber
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
RSC/N9501.Prepare for Rubber Mixing Operations	40	60	0	0	100	20
RSC/N9502.Load Raw Materials, Operate and Control Automated Internal Mixer	40	60	0	0	100	40
RSC/N9503.Monitor Product and Process Quality During Mixing	40	60	0	0	100	25
RSC/N5007.Follow health & safety protocol	30	70	-	-	100	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	170	280	-	-	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.